

AGGREGATION-LEVEL SEMANTIC MISMATCH: ROOT CAUSE ANALYSIS

Issue Summary

Benchmark energy (23.7796 mWh for 30 runs) is being compared against per-inference prediction (9.46-10.29 mWh), causing false budget exceeded events.

Critical Paths to Trace

Path 1: Benchmark Energy Accumulation

1. **Benchmark Start** → `/benchmark` endpoint called
2. **Meter Baseline Setup** → `baseline_energy_wh` set when FNB58 first connects
3. **30 Inferences Run** → Each inference runs, FNB58 accumulates power consumption
4. **Session Energy Computed** → `session_energy_mwh = (total_energy_wh - baseline_energy_wh) * 1000`
5. **Stored in** → `meter["measured_energy_mwh"]` and `meter["session_energy_mwh"]`

Result: `measured_energy_mwh` = 23.7796 mWh (TOTAL for all 30 runs)

Path 2: Per-Inference Prediction

1. **Model Deploy** → `prediction_energy_mwh` = 9.46-10.29 mWh
2. **Budget Set** → `energy_budget_mwh` = per-inference budget
3. **Telemetry Arrives** → Delta or cumulative readings from FNB58

Expected: Compare 9.46-10.29 mWh against budget **Actual:** ???

Path 3: Budget Comparison Logic

Location: `record_energy_sample()` in `server.py` line 679

```
over_budget = budget is not None and sample["energy_mwh"] > budget
```

Question: What is passed as `energy_mwh` when?

- During `/predict` → per-inference delta (correct)
- During `/benchmark` → ???
- From `/telemetry` delta payload → depends on payload source

Path 4: `measured_energy_mwh` Usage

Stored at: `server.py` line 943

```
meter["measured_energy_mwh"] = round(session_energy_mwh, 4)
```

Semantic: This is SESSION cumulative, NOT per-inference

Problem: Is this value ever passed to budget checks?

REQUIRED DEBUG OUTPUTS

Query 1: Where is latest_inference_delta_mwh set?

- Answer: Line 1098 in server.py during `end_inference_measurement()`
- Source: Computed from FNB58 meter deltas (should be per-inference)

Query 2: When does measured_energy_mwh get populated?

- Answer: Line 943 in server.py during meter snapshot updates
- Source: $(\text{total_energy_wh} - \text{baseline_energy_wh}) * 1000$ (SESSION cumulative)

Query 3: Where is measured_energy_mwh used in budget checks?

- Search needed: Look for direct usage of measured_energy_mwh in budget logic

Query 4: What telemetry payload arrives during benchmark?

- Does FNB58 send ONE big delta for all 30 runs?
- Does it send 30 small deltas?
- Is there a telemetry window mismatch?

Query 5: What gets stored in energy_metrics.history?

- Is per-inference delta recorded?
- Or is session total recorded?
- When is `record_energy_sample()` called during benchmark?

HYPOTHESIS

The benchmark path likely does NOT call `start_inference_measurement()` / `end_inference_measurement()` for each run.

If benchmark just runs 30 inferences without per-inference measurement wrapping:

1. No `latest_inference_delta_mwh` is set
2. Only `measured_energy_mwh` (session cumulative) gets updated
3. If telemetry arrives AFTER benchmark completes, a single "delta" payload might contain the TOTAL benchmark energy
4. This total is compared against per-inference budget → FALSE "over_budget"

REQUIRED INVESTIGATION STEPS

1. Check if benchmark_loaded_model calls measurement functions
2. Check if FNB58 telemetry is being batched into single large payloads
3. Check if record_energy_sample is being called with aggregated energy values
4. Verify budget comparison uses LATEST INFERENCE DELTA, not measured_energy_mwh
5. Check UI rendering logic - is it comparing correct semantics?

EXPECTED FIX FLOW

1. Separate semantics explicitly:

- `benchmark_total_energy_mwh` = sum of all runs
- `benchmark_avg_energy_mwh` = total / runs
- `latest_inference_delta_mwh` = last single inference

2. Budget comparison uses:

- `benchmark_avg_energy_mwh` (NOT `benchmark_total_energy_mwh`)

3. UI displays separately:

- "Benchmark Total Energy: XX mWh"
- "Average Per-Inference: XX mWh"
- "Latest Inference: XX mWh"
- "Session Cumulative: XX mWh"